Smart Rice Field Intelligence System Using an Autonomous Ground Rover

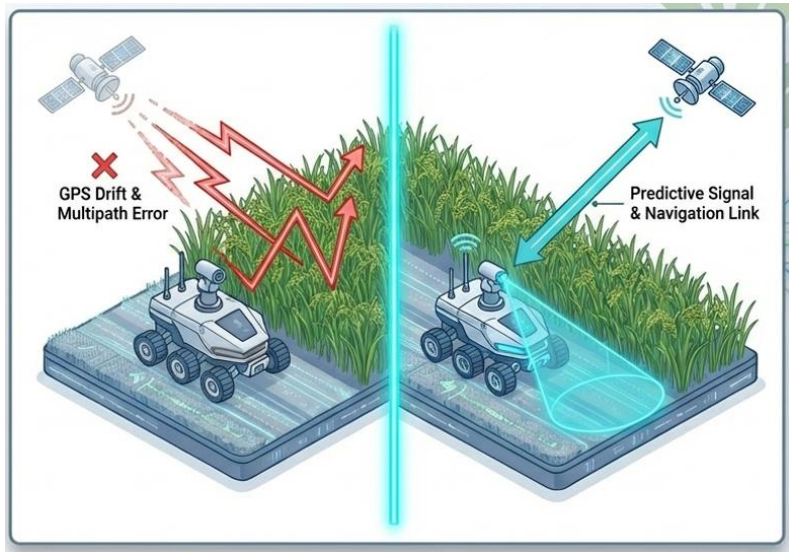
R26-IT-075



Introduction

- Rice farmers still depend on manual field inspections to monitor crop health and soil conditions, which is slow and limited in coverage
- Delayed detection of crop stress, disease, or soil problems leads to poor decisions on irrigation, fertilizer use, and treatment
- The main stakeholders are rice farmers, agronomists, agricultural officers, and farm managers who need accurate field information
- Existing methods rely on isolated tools and manual observation, which do not provide a unified and timely view of field conditions
- The proposed Smart Rice Field Intelligence System uses an autonomous ground rover to collect field data and support better farming decisions


Rice is Asia's staple crop, providing food
HeyGen





IT22112682-ARACHCI B N S

GPS Degradation Monitoring and Prediction Framework

Provides reliable positioning data for the rover by continuously monitoring GPS signal quality in rice field environments

Introduces a machine learning based framework to predict GPS degradation before it occurs

GPS DEGRADATION MONITORING AND PREDICTION FRAMEWORK

Problem Definition

• Autonomous agricultural systems depend on accurate GPS positioning for navigation and field data collection
• In rice field environments GPS signals often degrade due to weather conditions, terrain obstacles, and satellite signal instability
• GPS degradation can cause navigation errors, inaccurate field mapping, and unreliable data collection by the rover

Research Objective

• Develop a monitoring and prediction framework to detect and forecast GPS signal degradation in rice field environments for autonomous rover operations

Sub Objectives

• Monitor real time GPS indicators such as satellite count, HDOP, and signal stability during rover operation
• Identify patterns that indicate GPS signal degradation in agricultural field conditions
• Develop a machine learning based model to predict possible GPS degradation events
• Provide early alerts to the system to prevent navigation errors and maintain reliable field data collection

GPS DEGRADATION MONITORING AND PREDICTION FRAMEWORK

Research Methodology and Domain Contribution

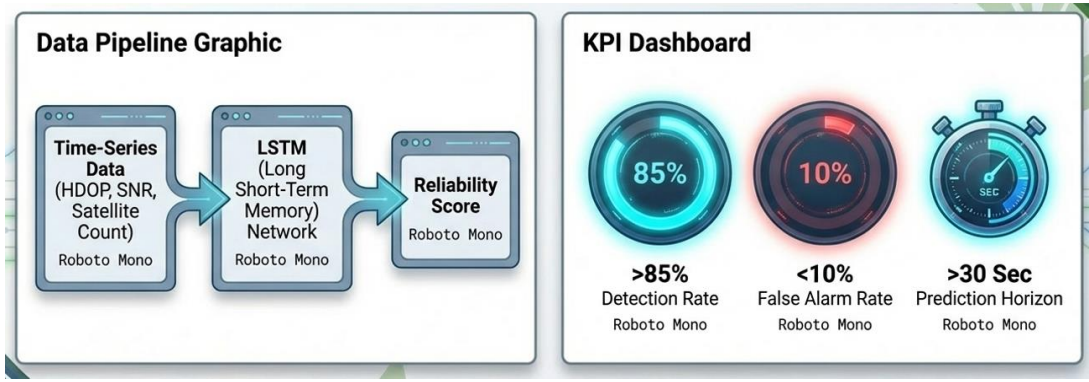
• Domain used: GNSS based positioning, autonomous navigation, time series monitoring, and machine learning

• The study monitors GPS quality indicators such as satellite count, HDOP, signal consistency, drift, and coordinate variation

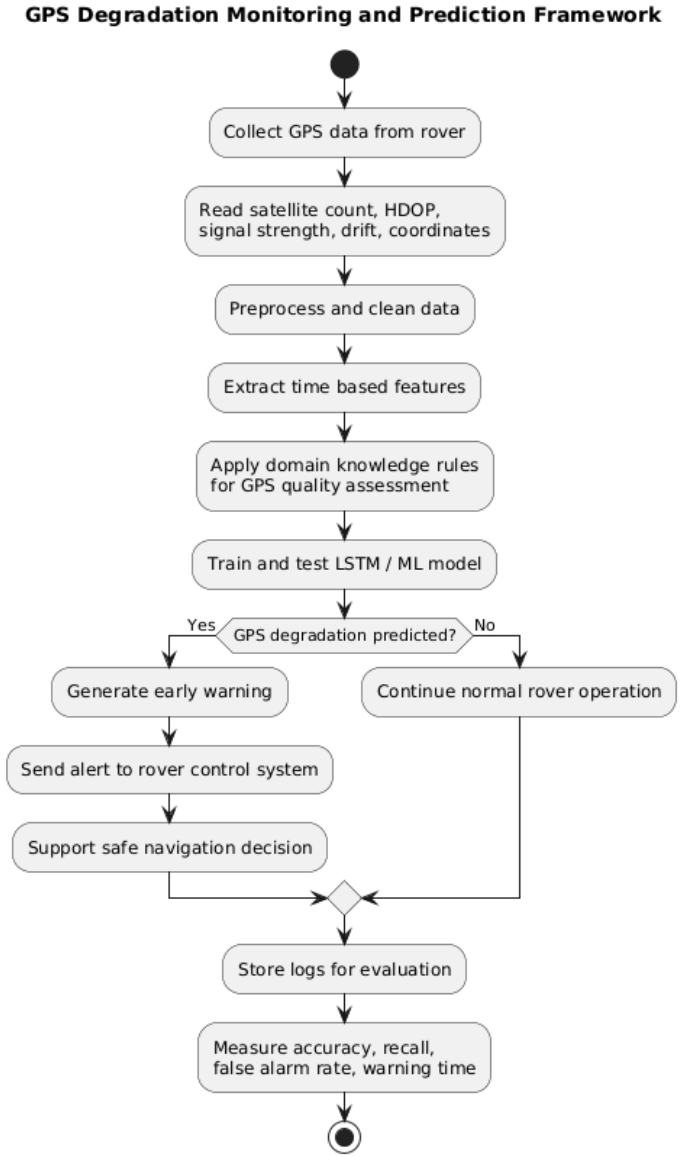
• These indicators are analysed to identify patterns linked with GPS degradation in rice field conditions

• A prediction model is then trained to detect degradation early and support safe rover movement

• This domain knowledge helps build a system that can reduce navigation errors and improve location reliability



GPS DEGRADATION MONITORING AND PREDICTION FRAMEWORK



GPS DEGRADATION MONITORING AND PREDICTION FRAMEWORK

System Overview of the Individual Component

- The rover collects GPS signals during movement across the rice field
- GPS indicators such as satellite count, HDOP, signal stability, and location drift are recorded continuously
- The collected data is preprocessed and analysed to identify patterns related to signal degradation
- A machine learning model processes the time series data to predict potential GPS degradation events
- When degradation is predicted, the system generates an alert to maintain reliable navigation and data collection

Dependencies and Common Components

- GPS/GNSS sensor module for real time positioning data collection
- Autonomous ground rover platform for field level data gathering
- Data processing environment using Python and machine learning libraries
- Communication module to transmit rover data to the processing system
- Shared system components such as central data storage, processing unit, and monitoring dashboard used by other project modules





IT22108722-KULASEKARA D K M

AI-Based Rice Crop Disease, Stress and Growth Monitoring

Detects rice crop diseases, plant stress, and growth stages using rover captured field images

Applies visual explanation methods to highlight affected plant areas

AI-BASED RICE CROP DISEASE, STRESS AND GROWTH MONITORING

Problem Definition

- Rice crops are vulnerable to diseases, nutrient stress, and environmental conditions that reduce yield
- Farmers often detect these problems only after visible damage appears
- Manual inspection of large rice fields is time consuming and often inaccurate
- Early detection of disease and stress is critical to prevent crop loss
- Current monitoring approaches do not provide automated field level crop health analysis

Research Objective

- Develop an AI based system that detects rice crop disease, stress conditions, and growth stages using field images captured by the rover

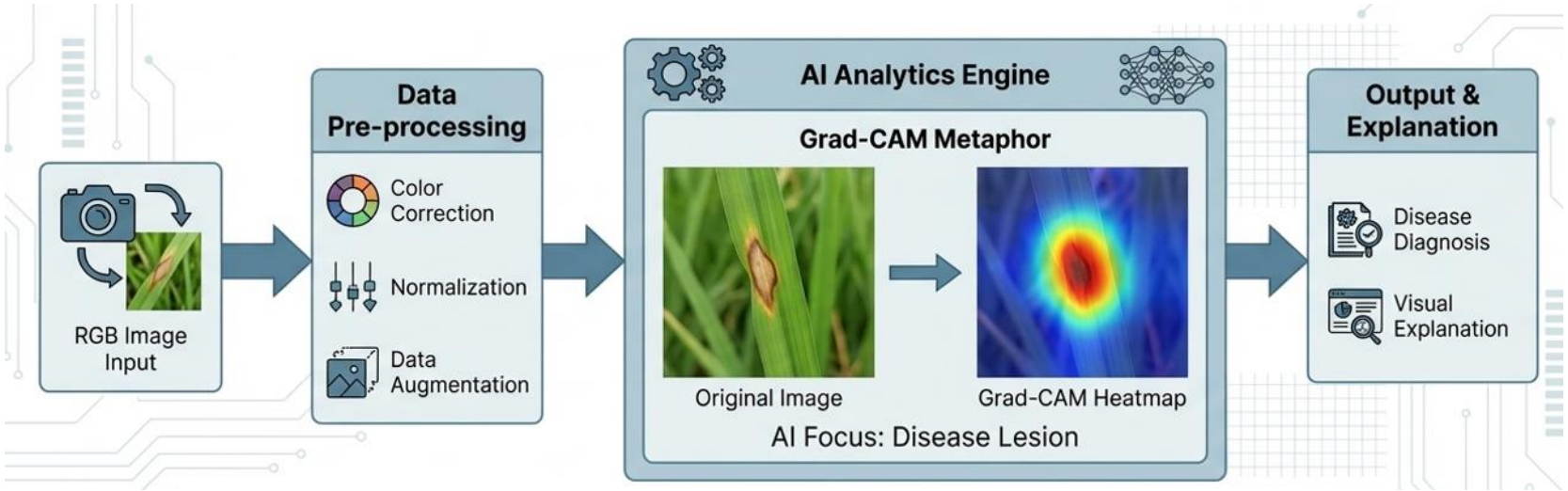
Sub Objectives

- Collect and preprocess rice crop images from field environments
- Train deep learning models to identify diseases and stress symptoms
- Detect different rice growth stages from plant images
- Provide visual explanation of model predictions to support decision making

AI-BASED RICE CROP DISEASE, STRESS AND GROWTH MONITORING

Research Methodology and Domain Contribution

- Domain areas include computer vision, deep learning, and precision agriculture
- Image datasets of rice crops are collected from field environments
- Images are cleaned, labelled, and prepared for model training
- Deep learning models analyse plant features such as leaf color, shape, and texture
- The trained model detects disease patterns, stress indicators, and growth stage characteristics

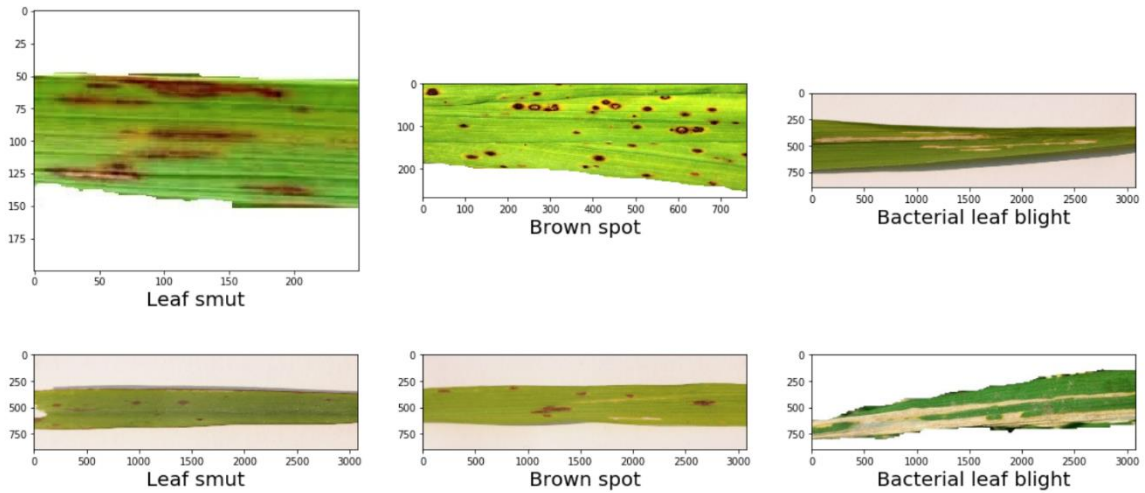


AI-BASED RICE CROP DISEASE, STRESS AND GROWTH MONITORING

Validation and Evaluation

- Evaluate the model using labelled rice crop image datasets
- Compare predicted results with expert labelled crop conditions
- Measure performance using accuracy, precision, recall, and F1 score
- Test the system with field images captured by the rover
- Validate whether the system can detect disease and stress early in real conditions

Data Availability



The Rice Leaf Diseases Detection dataset, contains a total of 3829 images, with each class containing approximately 600-plus images. The dataset includes the following disease classes: Bacterial Leaf Blight, Brown Spot, Healthy Rice Leaf, Leaf Blast, Leaf scald, and Sheath Blight.
https://www.kaggle.com/datasets/anshulm257/rice-disease-dataset

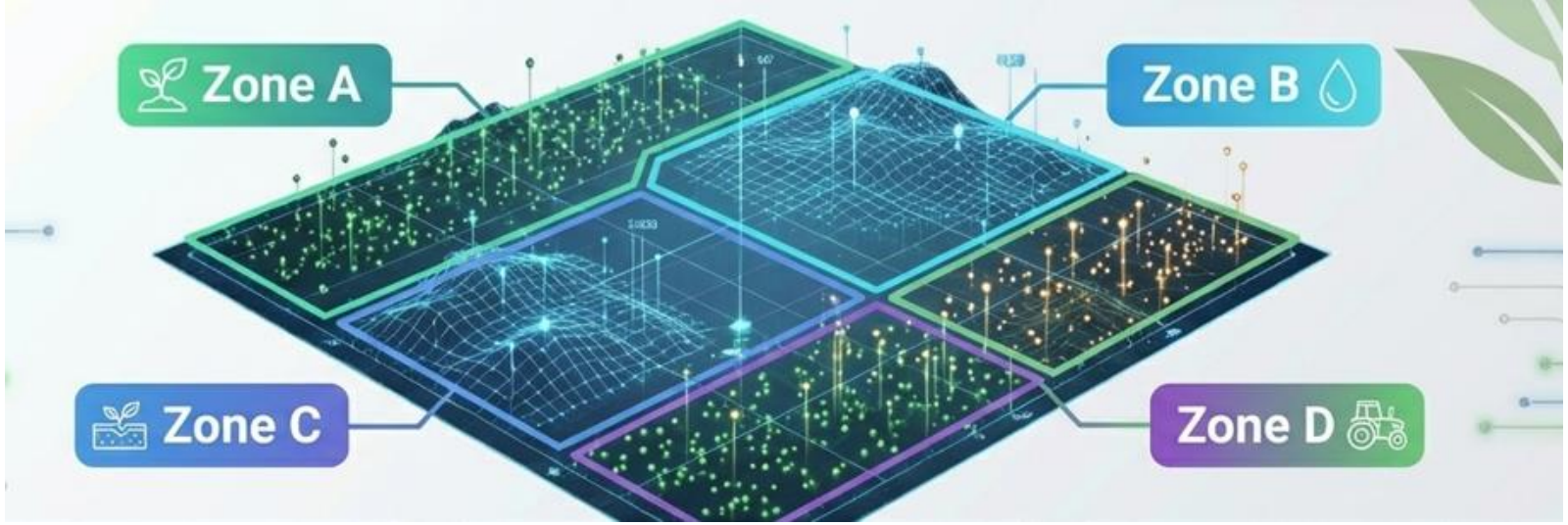
AI-BASED RICE CROP DISEASE, STRESS AND GROWTH MONITORING

System Overview of the Component

- The rover captures images of rice plants during field navigation
- Images are transmitted to the processing system for analysis
- Image preprocessing prepares the data for AI model prediction
- The deep learning model detects disease, stress conditions, and growth stages
- Results are sent to the central system to support field analysis and decision making

Dependencies and Common Components

- Camera module mounted on the rover for image capture
- Image processing and deep learning environment using Python
- Central data storage shared with other project components
- Communication module to transfer field images and results
- Integration with zone planning and soil analysis components for full system operation





IT22276032-S G M B SUBASINGHE

Intelligent Variable-Rate Fertilizer and Seeding Mission Control

Detects rice crop diseases, plant stress, and growth stages using rover captured field images

Introduces zone based field management using crop and soil data

INTELLIGENT VARIABLE-RATE FERTILIZER AND SEEDING MISSION CONTROL

Problem Definition

- Farmers usually apply fertilizer and seeds uniformly across the entire field
- Different areas of the rice field often have different soil fertility and crop conditions
- Uniform input application leads to inefficient fertilizer use and uneven crop growth
- Lack of data driven field management reduces productivity and increases cost
- A system is needed to generate precise fertilizer and seeding plans based on field data

Research Objective

- Develop an intelligent mission control system that generates variable rate fertilizer and seeding plans using crop health and soil condition data

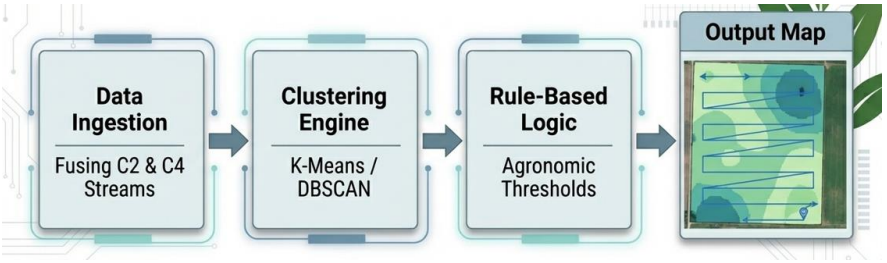
Sub Objectives

- Divide the rice field into management zones based on crop and soil information
- Analyse crop health and soil fertility data collected from other system modules
- Generate variable rate fertilizer and seeding recommendations for each zone
- Produce rover mission maps that guide fertilizer and seeding operations

INTELLIGENT VARIABLE-RATE FERTILIZER AND SEEDING MISSION CONTROL

Research Methodology and Domain Contribution

- Domain areas include precision agriculture, data driven farming, and geospatial analysis
- Crop health data from the monitoring component and soil data from the soil analysis component are used as inputs
- Field data is analysed to identify spatial variation across the rice field
- Clustering methods divide the field into zones with similar conditions
- Each zone receives specific fertilizer and seeding recommendations based on data patterns



Latest Technologies Used

- Machine learning clustering methods such as K means and DBSCAN
- Geospatial data analysis for field zone identification
- Python based data processing and analysis tools
- Mission planning algorithms to generate rover operation paths
- Data visualisation tools for displaying field zones and treatment plans

INTELLIGENT VARIABLE-RATE FERTILIZER AND SEEDING MISSION CONTROL

Validation and Evaluation

- Validate the system using real field datasets containing crop and soil information
- Compare generated fertilizer and seeding plans with expert recommendations
- Evaluate the accuracy of zone classification and input recommendations
- Measure performance using zone classification accuracy and recommendation effectiveness
- Assess whether the system improves resource use and field productivity

Data Availability, Security, and Ethical Considerations

- Data includes crop health data, soil condition data, and field location information
- Data collected from rover sensors and other project modules
- Data does not contain personal or sensitive information
- Stored securely and used only for research purposes
- System operation does not cause environmental harm when used correctly

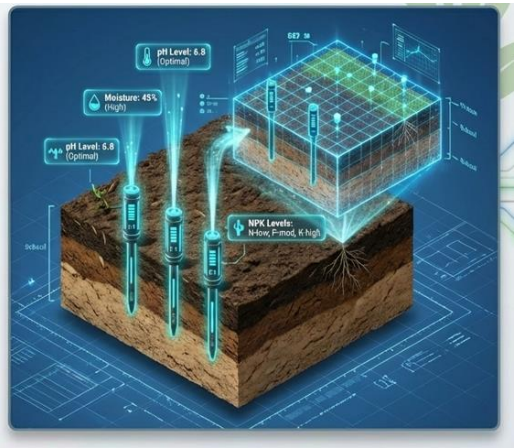
INTELLIGENT VARIABLE-RATE FERTILIZER AND SEEDING MISSION CONTROL

System Overview of the Component

- The system receives crop health data from the crop monitoring component
- Soil condition data is received from the soil analysis component
- Data processing identifies spatial variation across the rice field
- Clustering methods divide the field into management zones
- The system generates fertilizer and seeding mission plans for rover based field operations

Dependencies and Common Components

- Crop health monitoring module for plant condition data
- Soil analysis module for soil fertility and nutrient information
- Field location data generated using the rover navigation system
- Central data processing platform shared with other components
- Rover control system to execute fertilizer and seeding missions





IT22192950-S.N.R JAYAWARDHANA

Predictive Soil Intelligence and Rice Field Decision Support

Supplies soil condition data to the variable rate fertilizer and seeding module

Detects early signs of soil nutrient deficiency and water stress

PREDICTIVE SOIL INTELLIGENCE AND RICE FIELD DECISION SUPPORT

Problem Definition

- Soil fertility and moisture conditions vary across different areas of a rice field
- Farmers often rely on manual observation and general recommendations for soil management
- Lack of precise soil information can lead to improper fertilizer and irrigation decisions
- Poor soil management reduces crop productivity and increases resource waste
- A data driven system is needed to monitor soil conditions and guide better farming decisions

Research Objective

- Develop a predictive soil intelligence system that analyses soil conditions and provides decision support for rice field management

Sub Objectives

- Collect soil sensor data such as moisture, temperature, pH, and nutrient levels
- Analyse soil condition patterns across the rice field
- Develop predictive models to estimate soil health and nutrient status
- Provide recommendations for irrigation, fertilizer application, and soil improvement

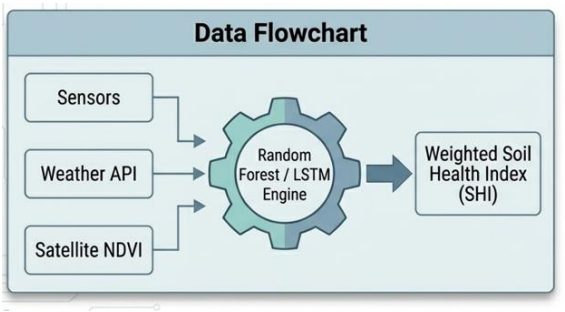
PREDICTIVE SOIL INTELLIGENCE AND RICE FIELD DECISION SUPPORT

Research Methodology and Domain Contribution

- Domain areas include soil science, environmental monitoring, and precision agriculture
- Soil data is collected using sensors installed on the ground rover platform
- Data preprocessing prepares soil measurements for analysis

Latest Technologies Used

- Soil sensor modules for pH, moisture, temperature, and nutrient measurement
- Machine learning models for soil condition prediction
- Python based data analysis and model training tools





PREDICTIVE SOIL INTELLIGENCE AND RICE FIELD DECISION SUPPORT

Validation and Evaluation

- Validate predictions using soil data collected from real rice field environments
- Compare predicted soil conditions with actual sensor measurements and expert assessments
- Evaluate prediction performance using accuracy, precision, recall, and prediction error metrics
- Assess whether the system improves irrigation and fertilizer decision making
- Measure the reliability of soil recommendations across different field zones

Data Availability, Security, and Ethical Considerations

- Soil sensor data collected from the rover platform and field monitoring activities
- Data includes environmental and soil measurements without personal information
- Data will be stored securely and accessed only for research purposes
- Field testing will be conducted without damaging crops or soil resources
- Ethical risk is minimal because the study focuses on environmental data analysis

PREDICTIVE SOIL INTELLIGENCE AND RICE FIELD DECISION SUPPORT

System Overview of the Component

- The rover collects soil measurements using sensor modules while moving through the field
- Soil data is transmitted to the processing system for analysis
- Data preprocessing prepares the information for prediction models
- Machine learning models estimate soil health and nutrient status
- The system generates recommendations for irrigation, fertilizer use, and soil management

Dependencies and Common Components

- Soil sensor modules mounted on the rover platform
- Rover navigation system for field level data collection
- Data processing environment for machine learning analysis
- Central database shared with other system components
- Integration with crop monitoring and mission planning modules to support full system decision making

Smart Rice Field Intelligence System Using an Autonomous Ground Rover

Value Proposition



Protects up to 15% yield loss and reduces chemical input costs by 20-35% via targeted application. Makes precision-ag affordable without drone licensing.

Target Market



Smallholder farmers, Ag-cooperatives, and Government Extension Services across South/Southeast Asia.

Revenue Model



B2C Freemium mobile app (LKR 500-800/mo premium), B2B Institutional licensing, and B2B Data Services.

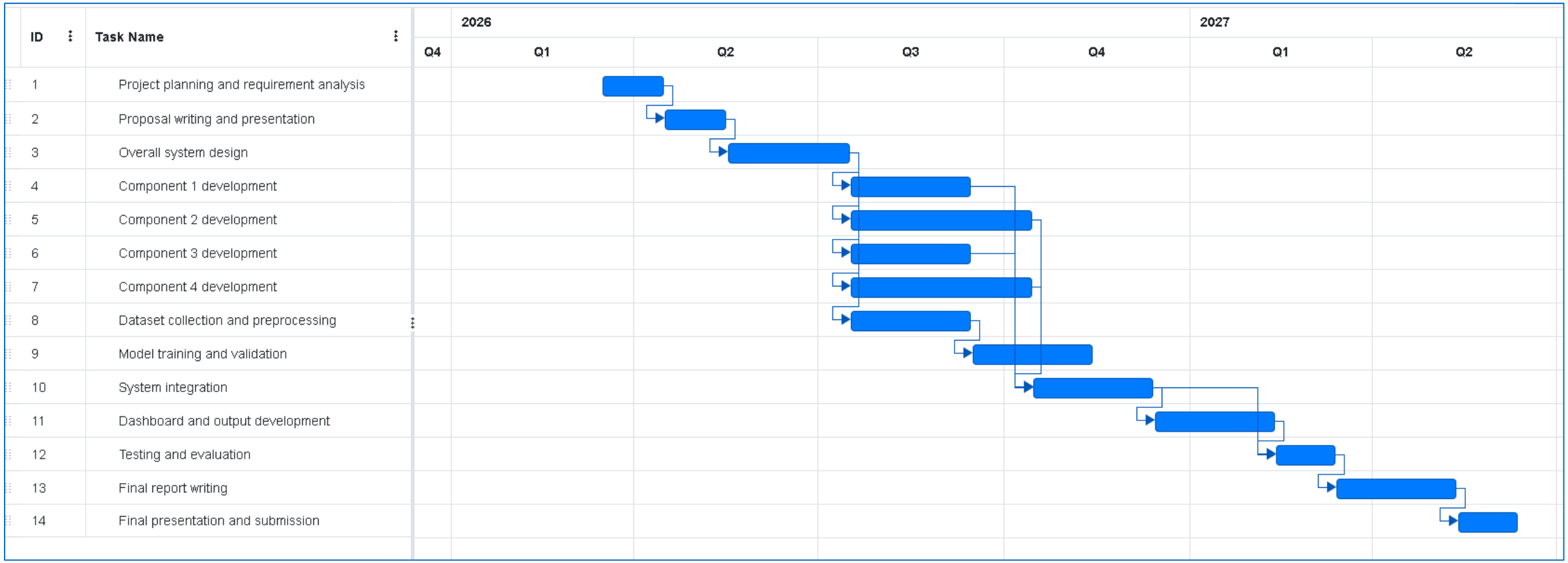
IP & Costs



Software copyrights, Grad-CAM/Soil algorithmic fusions. Estimated Initial R&D Cost: ~LKR 300K - 450K.

Smart Rice Field Intelligence System Using an Autonomous Ground Rover

Gantt Chart



Smart Rice Field Intelligence System Using an Autonomous Ground Rover

Work Breakdown Structure

